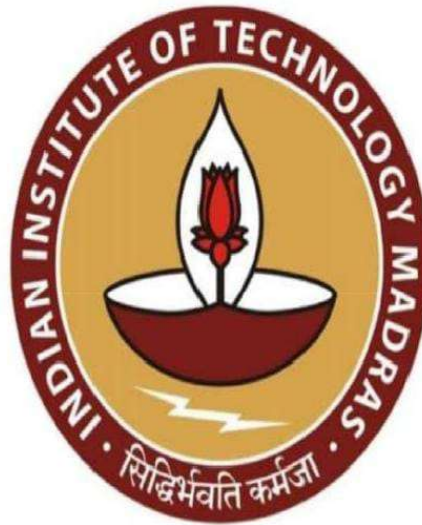


Developing a comprehensive business strategy for effective business growth

A Mid-term report for the BDM capstone Project

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Table of contents

Content	Page no
1.Executive Summary	3
2.Proof of Originality	4
3.Meta data and Descriptive Statistics	5 - 9
4.Detailed explanation of Analysis method/process	10 - 11
5.Results and Findings	12 - 14

EXECUTIVE SUMMARY

This mid-term report provides a comprehensive financial and operational analysis of Veeraih and Co., a cement dealership in Guntur, Andhra Pradesh. Established in 2024, the business operates with five employees and generates an annual income of ₹12 lakhs (estimated) as it is a new business which is producing an average income of 1 lakh rupees per month. The main problems faced by the business which were identified in conversation with the owner are:

1. High Transportation Costs
2. Low Profit Margins
3. Labour Shortage

To deal with the problems mentioned above, the main approach that is followed is:

1. Identify the problem
2. Analyse the root causes
3. Collect the data
4. Analyse the data
5. Develop the solution and implement

The first two steps has been completed in the proposal report and now in this report I will be working on the 3, 4 steps.

Firstly, raw data has been collected from the owner and it was organized and then converted it to a dataset and summarization has been done. (Calculation of totals, averages etc and creation of graphs for analysis)

The variables used in the dataset were described in Meta data section and a detailed analysis on the dataset was done in descriptive statistics section and the insights obtained from it were mentioned under that section.

Also the detailed explanation of the analysis methods used and process was done with all the useful information that has been gathered.

The main goal of this report is to identify the key operational and financial challenges faced by the cement dealership business and increase the business profitability by 25 – 30 percent. Methods and processes that were used to increase the profitability has been explained in this report clearly.

PROOF OF ORIGINALITY

INTERACTION VIDEO LINK:

<https://drive.google.com/file/d/1eL0HpHVADPXLZJreYJ79G-4wRKuC2saQ/view?usp=drivesdk>

TRANSCRIBED VIDEO TO TEXT LINK:

<https://drive.google.com/file/d/1emGvJsyJtaerhWN88DKhdrFrxxmnYBkda/view?usp=drivesdk>

IMAGES AND PROOF OF ORIGINALITY LETTER LINK:

<https://drive.google.com/drive/folders/17QZYPFMYEAuKkzn1KTypZdG-yUphIIT7>

META DATA

This section describes the dataset used for analysis, detailing the data collection process, time coverage, source, and key variables.

Data collection process:

- Source: From the discussion with owner, the data has been collected and has been written manually into the excel.
- Data type: Monthly operational and financial data covering August 2024 to March 2025.
- Collection Method:

Manually compiled from business invoices, supplier receipts, and transport records. Cross- verified with the owner of the business.

- Frequency: Data is monthly, with one record per month representing aggregate figures for key business variables.
- Time period covered :

Start time: August 2024

End Time: March 2025

Duration : 8 months

BUSINESS OR FINANCIAL META DATA

VARIABLE	TYPE	UNIT	DESCRIPTION
SALES REVENUE	Numeric	₹	Total amount of money generated before subtracting any costs.
OPERATIONAL COST	Numeric	₹	Overheads such as rents, wages, electricity, initial investment and etc
TRANSPORTATION COST	Numeric	₹	Costs due to transportation of cement.
TOTAL COST	Numeric	₹	Sum of transportation and operational costs.
PROFIT	Numeric	₹	Revenue minus total cost
PROFIT MARGIN	Numeric	₹	Profit as a percentage of sales revenue
CASH FLOW	Numeric	₹	In flow and out flow of money in a given month

OPERATIONAL META DATA:

VARIABLE	TYPE	UNIT	DESCRIPTION
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MONTH	Categorical	N/A	Calendar month of the year.
PEAK SEASON INDICATOR	Categorical	Month	Months when sales was highest.
DELIVERY TIME	Numeric	Days	Average time taken for the cement to reach from supplier to dealership.

PRODUCT META DATA:

VARIABLE	TYPE	UNIT	DESCRIPTION
Sales Volume	Numeric	Tons	Quantity of cement sold monthly.
Stock before sales	Numeric	Bags	Inventory available at the start of the month.
Stock after sales	Numeric	Bags	Inventory remaining at the end of the month.

DESCRIPTIVE STATISTICS

The values are obtained by using excel formulas like SUM, MEDIAN, STANDARD DEVIATION, MAX, MIN and formulas to calculate the below mentioned terms is written,

1. Revenue = bags sold*₹310 (average cost)
2. Total cost = Operational cost + Transportation cost
3. Operational cost = stock purchase cost +rents +wages +utilities cost(like electricity,water,etc)+marketing
4. Profit = Revenue – Total cost
5. Profit Margin = (Profit/Revenue)*100
6. Stock before sales = stock after sales (last month)+Monthly production(monthly production of bags was 10,000 bags)
7. Stock after sales = Stock before sales +Production – bags sold

Variable	Mean	Median	Standard Deviation	Max	Min
Bags sold	10,100	10,000	614.4102864	11,200	9,200
Revenue(₹)	31,23,250	29,91,500	198950.2136	34,72,000	28,52,000
Transportation cost(₹)	1,01,000	1,00,000	6144.102864	1,12,000	92,000
Operational cost(₹)	29,32,500	29,00,000	194213.1561	32,80,000	26,50,000
Total cost(₹)	30,33,500	30,00,000	200353.4377	33,92,000	27,42,000
Profit(₹)	97,500	1,00,000	11989.5788	1,10,000	80,000

Profit margin(%)	3.14	3.26	0.54	3.86	2.30
Stock before sales(bags)	599	300	621	1800	30
Stock after sales(bags)	376	150	444.7453625	1300	20

INSIGHTS OBTAINED FROM ABOVE STATISTICS

1. The average no of bags sold per month is 10100, as per the data collected from the owner, each bag weighs 50kg, that is the average no of tonnes sold per month is 500 tonnes.
2. As per the data collected from the owner, each bag of cement (50kg weight) costs on an around 310 rupees, on analysing the above data, the profit earned by the owner on each bag is around 9.75 rupees(approx) which is obtained by calculating average profit/average no of bags sold.
3. The revenue's standard deviation shows moderate variability (6 percent of average revenue) indicates fluctuations in sales volume likely due to seasonal or demand-based factors.(like the months of November, December, January where the demand is high).
4. The data analysis above shows low profits and profit margins.
5. The sales fluctuation from 9,200 bags to 11,200 bags shows variation among months, including peak and low-demand periods.
6. The average profit margin is 3.14% which indicates low-margin, high volume business.
7. The max capacity of storage of business is 1800 bags.
8. The profit per bag is 9.75 rupees is quite low.

DETAILED EXPLANATION OF ANALYSIS METHODS /PROCESS:

The data has been collected using time series analysis technique for a time period of 8 months (August, 2024 to March, 2025). This analysis is used as the data has been collected over a certain duration of time in regular intervals.

Mainly, the data has been analysed using three different types of analysing techniques, namely:

1. Descriptive Analysis
2. Diagnostic Analysis
3. Trend Analysis

The process of collecting data:

There was no written or digital storage of data, the only means to collect the data is from the discussion with the owner, write it manually into excel and perform all the required analysis. With the help of excel the above descriptive analysis has been done. Now moving on to the analysis process,

1. DESCRIPTIVE ANALYSIS

The basic descriptive analysis of the data has been mentioned above. It provides a statistical summary to understand what is happening in the data. This analysis helps to answer the question “what happened?”.

Specific objectives of descriptive analysis:

- Understand the business performance.
- Determine if costs are aligned with production/sales volume.
- Analyse profit margins and monthly earnings.
- Highlight areas of improvement or optimization.

2. DIAGNOSTIC ANALYSIS

This analysis helped to investigate the causes behind patterns and deviations found in the data.

The insights obtained from the diagnostic analysis are

- Profit changes month to month even though the selling price is fixed, root cause of this problem is changes in bags sold and operational cost.
- Months with higher profit has higher sales or lower operating costs.
- The main reasons that drives operational cost variations are seasonal energy, labour or raw material cost changes, overproduction during off seasons(ex: march 2025).
- High fixed operational costs in proportion to unit price has led to low profit margins, to improve margins either cost price needs to be dropped or selling price must increase.
- Production is not aligned with the sales, sometimes there is overstocking, which is over production in off season like the month of August, September.

3. TREND ANALYSIS

This analysis helped to focus on identifying consistent patterns over time and helped to understand how key business metrics are evolving month by month.

Insights developed from this analysis are:

- Sales range from 9200 to 11200 bags/month. One month showed spike (high sales) and other dips aligning with typical seasonal trends (as shown in excel the months with high sales is November, December and January and low sales is August and September).
- Sales volume is the primary driver of revenue as there is no change in pricing strategy.
- Operational cost rises and falls with sales/production volume, i.e more production = more cost.

THE DETAILED ANALYSIS OF TRENDS HAS BEEN DONE IN RESULTS AND FINDINGS SECTION.

RESULTS AND FINDINGS

The insights drawn from the analysis of the data



Chart -1

- The chart above shows the monthly sales of cement bags for a interval of 8 months. This helps to understand if there is any trend in sales (i.e if there is any seasonal demand for the stock).
- This analysis helps to prevent overstocking or stockouts which helps to reduce wastage in production cost (which is part of operational cost).
- From the chart above we can see that the sales were high in the months from October to February (Post monsoon months & winter) as they are ideal months for construction. The government projects usually ramp up and real estate demand spike.
- The peak months were November, December, January
- The months where the demand is low as seen from the above chart is August, September. Usually construction slows down due to rain and flooding. Transportation issues also delay supply in these months.

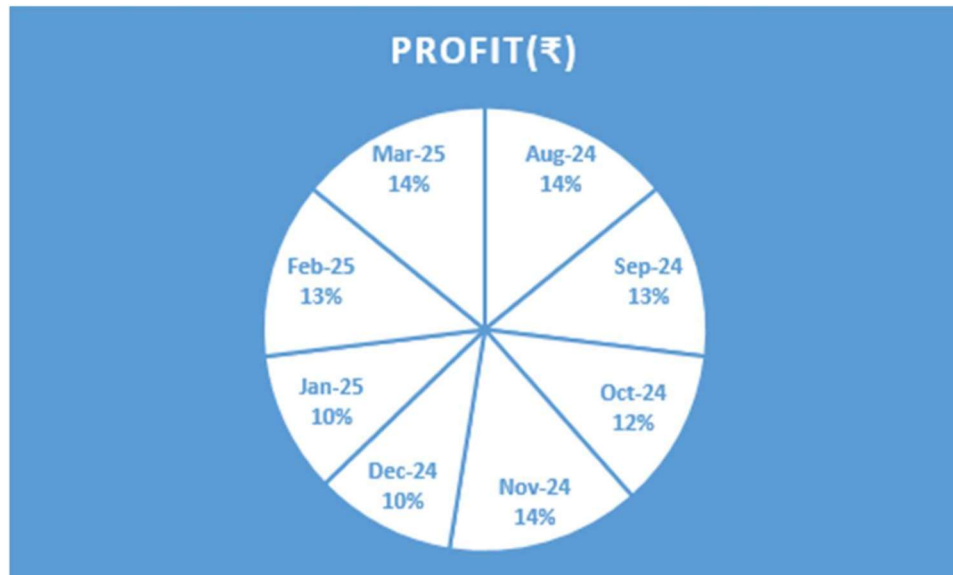


Chart 2

- The above chart is created to find out if there is any relation between the sales volume and the profit. From the chart 1, the peak months were found to be November, December, January but from chart 2, the profit is high in the months November, August, March. So, from both chart 1, chart 2 it can be inferred that the months with high sales is not related to the months with high profits.
- Even when there is high sales volume, there is no high profit that might be because of high transportation cost or high operational costs in that particular month.
- To reduce high transportation costs, the business can start using electric vehicles, which would cut down the costs drastically by 10 percent and additionally the government would provide subsidy on purchasing and registering the electric vehicle.
- The government subsidy adds profit to the business.

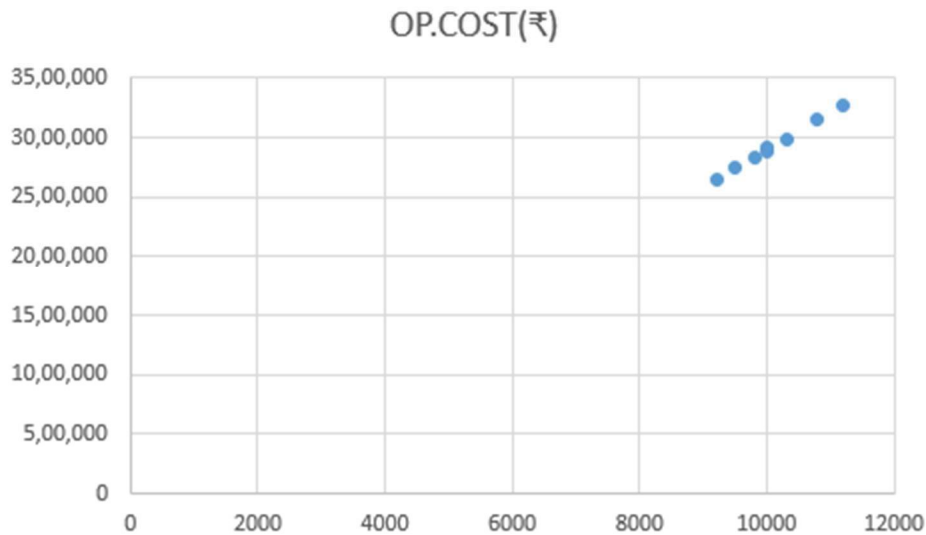


Chart 3

- The above scatter plot (chart 3) shows the relation between the operational cost and bags sold.
- As seen above the operational cost is increasing linearly with the no of bags sold. There are likely few fixed costs but most expenses like raw materials, labour, packaging etc increases as the production increases.
- Some ways to reduce the operational costs with increasing production is to use machinery instead of man power to reduce labour wages and errors, as the labour shortage is also one of the major problems faced by the business.
- Increase market reach by registering in online sites like IndiaMART, TradeIndia, Udaan and other sites. These sites help the owner to reach bulk buyers across India or even globally. Also verified listing on trusted platforms increases buyer confidence.